

“OpenVPCal”: An Open Source In-Camera Visual Effects Calibration Framework

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ABSTRACT

This presentation introduces the OpenVPCal toolset, an open source in-camera visual effects calibration framework. The toolset includes reference patch generation, creation of calibration color transforms, and custom 1D roll off lookup tables (LUTs) to control content brightness while attempting to maintain a linear light output from the LED panels. The resulting transforms can be expressed via an OpenColorIO Config [OCIO Contributors 2022] file, making tracking and transport easier for productions to manage and reproduce. We hope that by making this workflow open source and accessible, it will increase dialogue between practitioners in the space, leading to new novel improvements for creating more amazing content worldwide.

CCS CONCEPTS

• **Software and its engineering** → *Open source model*; • **Computing methodologies** → *Image-based rendering*.

ACM Reference Format:

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1 INTRODUCTION

In today’s production environment, LED panel based workflows often referred to as “in-camera visual effects” (ICVFX) have become more and more common. The issues associated with these workflows between LED panels and camera capture have also been well documented (moire, shutter sync, resolution, narrow-band emittance from LED panels and their limited imaging performances). One of the biggest challenges of correctly achieving ICVFX is to create a seamless integration between the physical scene and the digital scene. This integration needs to be valid from the camera observer, not from the human observer. This work focuses on a fully open source calibration framework - from content playback to LED panel to camera capture, designed to make the best use of today’s technology.

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2 DESIGN AND IMPLEMENTATION

OpenVPCal is designed to generate a set of color transforms that aim to calibrate the LED panel output to meet an intended target color space and its EOTF, while minimizing the metamerism errors generated by the combination of the narrow-band LED emittance and the camera observer used on set. The calibration is achieved by applying the color transforms to the content’s image pipeline that is projected on the LED panel and offsetting it to modify its color reproduction in order to appear correct when viewed through the camera. The tools are designed to work without requiring any additional calibration equipment such as a dedicated colorimeter or spectrometers. Furthermore, the calibration tool tries to overcome the most common complexities encountered in ICVFX calibration workflows while aiming for the process to be equally functional and take as little time as possible on-set. The color transforms are generated by analyzing the RAW image data produced by the on-set camera via recording specifically designed color patches and reference images which are projected on the LED panels.

2.1 Tool set and workflow

The framework is composed of several tools designed to work in a chain. The “Patches Generator”, the “Calibration Tool”, the “Calibration Verification Tool” and the “Parametric Roll-off Tool”. The user is required to first export the calibration patches via the Patches Generator, and load them into the media player of preference that is connected to the LED wall pipeline. Through a simple series of instructions, the user is guided through the steps to set up the camera to capture the calibration patches. Once shot, the footage is imported into the Calibration Tool. Once the quality of the captured calibration patches is determined, the tool analyzes the footage and generates the required calibration transforms.

2.2 Calibration transforms

The set of calibration transforms are 1) a 3x3 Matrix, 2) 3x 1D LUTs, 3) the ACES parametric gamut compression algorithm. The 3x3 matrix is utilized to convert the white point and the color gamut of the LED panel as seen by the camera to the target. The 3x 1D LUTs are used to balance the gray scale and correct the EOTF. The parametric ACES Gamut Compression [Shaw et al. 2022] is tuned to compress any out of gamut colors that the LED panel is unable to reproduce.

2.3 Roll-off

In addition to the calibration workflow, we developed a parametric roll-off calculator that generates a 1D LUT to be used in conjunction with the calibration transforms. While the calibration process aims to normalize the LED panel signal to meet the expected linear light

output, the roll off is required to tone map and manage the content's native brightness to best fit the LED panel's capabilities. Given the limited dynamic range/contrast ratio of the LED panel, the image pipeline requires a non-invasive tone-mapping that can compress only the image values outside the dynamic range of the LED panel, whilst keeping most of the rest of the image data untouched (i.e. linear). The proposed roll-off tool will use a number of parameters that can help the algorithm determine: a) the amount of image data to compress, b) the aggressiveness of the compression curve, c) what to clip out.

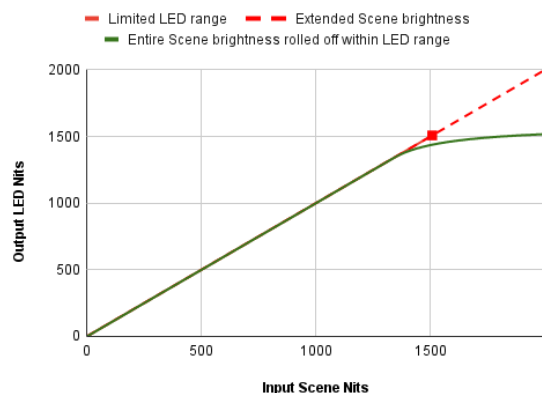


Figure 1: Example Roll-Off implementation.

2.4 OCIO Config and packaging

All these above-mentioned color transforms can be exported as a single 3D LUT or as independent transforms compatible with OCIO. As not everyone is familiar with OCIO configs, we also suggest a reference OCIO config with a pipeline that is designed to host the color transforms in the correct order without requiring manual setup.

3 CONCLUSION

The primary focus of this tool set is the on-set production process. We recognize that from a pure imaging and color science perspective, we have taken some “shortcuts” - a camera is not a colorimeter, and current LED panel technology should not be expected to accurately reproduce broad spectrum light/colors. However, it is important that we maintain a workflow to make the best of current technology while keeping the time impact on set as low as possible. It is crucial to democratize these workflows globally, which is why making these tool sets open source and collaborative matters. We look forward to a future with improved LED panel technology such that calibration workflows should be less necessary.

ACKNOWLEDGMENTS

Adrian Pueyo and the team at ORCA Studios, who are co-authors of this framework and were integral in developing the first version of this tool. Rod Bogart at Epic Games for his advice and feedback.

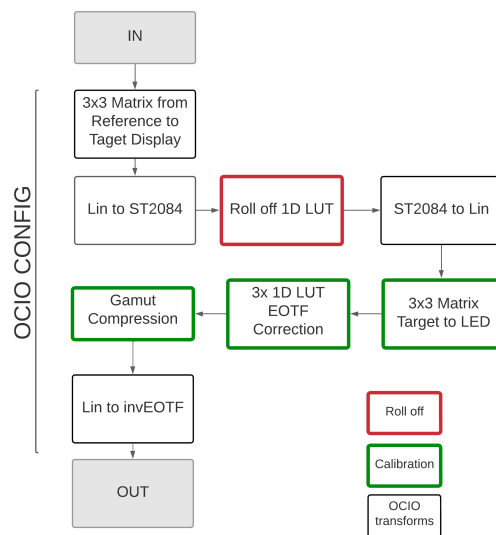


Figure 2: Example OCIO Config Structure.

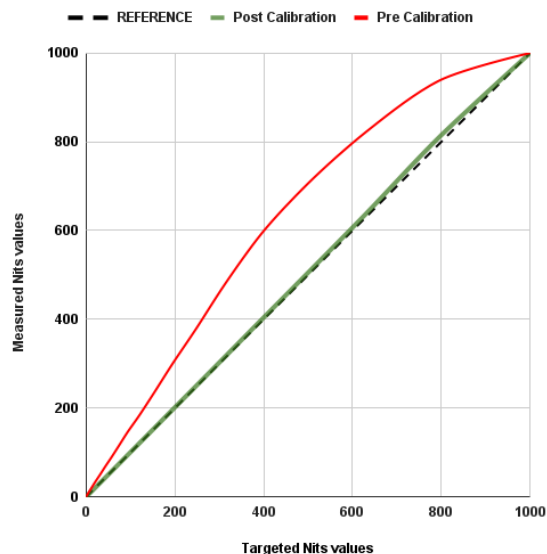


Figure 3: Pre vs. Post OpenVPCal Calibration.

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